



Rogue planets in space

Astronomers believe they have discovered at least 70 new free-floating exoplanets not tied to any stars. <https://digit.in/jan22-37>



Images of Mars

The ESA has released never-seen-before images of Mars. <https://digit.in/jan22-38>

Rocket fuel from Atmospheric CO₂

On the lookout for more sustainable ways to fuel our spaceflights

Harshita Arya | feedback@digit.in

I

t is a well known fact that the Earth is suffering from climate change. Carbon Dioxide

(CO₂) emissions have been rising rapidly ever since the industrial revolution in the early 20th century and global temperatures have been on the rise triggering a positive feedback cycle and threatening to make it worse. According to reports, even if industrialised nations agree to and seriously comply with slashing carbon emissions drastically, warming won't begin to slow until the mid-century. The emission reduction has to be paired with carbon capture (CC) in order to stand a chance at avoiding the worst-case eventualities.

Elon Musk, the chairman and the chief executive officer of Space Exploration Technologies Corp, more popularly known as

SpaceX, announced the Carbon Capture project on December 13 after being named the Person of the Year by Time magazine.

While Elon Musk believes that improving access to space is critical to our long-term survival, he also stated that the allure of outer space technologies and achievements steals focus from Earth's problems and that rockets produce excessive carbon emissions. Therefore, he declared that SpaceX will begin the Carbon Capture (CC) program to create propellants for his rockets.

According to a New York Times report, SpaceX currently uses

liquid oxygen with kerosene to power their Falcon rockets and have been launching satellites into orbit, delivering supplies and astronauts for NASA (National Aeronautics and Space Administration) and other customers.

WHAT IS CARBON CAPTURE AND HOW WILL IT WORK?

- To get started carbon dioxide needs to be removed from the atmosphere through air scrubbers, absorption, chemical catalysts or other methods.
- It is then combined with water, creating hydrocarbons which essentially forms the basis of fuels we already use like petroleum and natural gas. The process creates biofuel and this biofuel produces fewer emissions than the fossil fuels we use today.
- This basically results in negative emissions which not only reduces the level of CO₂ in



Image credit: New York Post



Image Source: News 18



Image Source: carbonquest